Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 11 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 11 : SLO â€“ 2 | Circular Queue â€” Enqueue Operation

I. Algorithm to Insert an Element in a Circular Queue

- Step 1: IF FRONT = 0 AND REAR = MAX-1 (OR FRONT = REAR + 1)
- Write OVERFLOW
- Goto Step 4
- [END OF IF]
- Step 2: IF FRONT = -1 AND REAR = -1
- SET FRONT = REAR = 0
- ELSE IF REAR = MAX-1 AND FRONT != 0
- SET REAR = 0
- ELSE
- SET REAR = REAR + 1
- [END OF IF]
- Step 3: SET QUEUE[REAR] = VAL
- Step 4: EXIT

II. Programming Code â€” insert()

- void insert() {
- int num;
- printf("\n Enter the number to be inserted in the queue : ");
- scanf("%d", &num);
- if ((front == 0 && rear == MAX-1) || (front == rear + 1))
- printf("\n OVERFLOW");
- else if (front == -1 && rear == -1) {
- front = rear = 0;
- queue[rear] = num;
- }
- else if (rear == MAX-1 && front != 0) {
- rear = 0;
- queue[rear] = num;
- }
- else {
- rear++;
- queue[rear] = num;
- }
- }